Module 8 Lesson 1
Basics of Hypothesis Testing (8.1)

Objectives

In this lesson, we will learn how to

- identify the null hypothesis and alternative hypothesis;
- understand significance level, test statistic, P-value, and critical region;
- make decisions using the P-value method and the critical value method;
- write final conclusions in simple language;
- describe Type I and Type II errors;
- understand the idea of power in a hypothesis test.

1. What Is a Hypothesis Test?

In statistics, a hypothesis is a claim or statement about a property of a population.

A hypothesis test is a formal procedure for testing a claim about a population parameter.

Main Idea

A hypothesis test uses sample data to decide whether there is enough evidence to reject a statement about a population.

Common population parameters include

p population proportion, μ population mean, σ population standard deviation.

2. Example of a Claim

Suppose someone claims:

Most Internet users utilize two-factor authentication to protect their online data.

Let

$p =$ the proportion of Internet users who utilize two-factor authentication.

The word ``most'' means more than half, so the claim can be written as

$$p > 0.5.$$

Question We Are Asking

Is the sample evidence strong enough to support the claim that

$$p > 0.5?$$

3. Null and Alternative Hypotheses

The null hypothesis, denoted by H_0, is a statement that a population parameter is equal to some claimed value.

The alternative hypothesis, denoted by H_1 or H_a, is a statement that the parameter differs from the null hypothesis.

Important Rule

For the methods in this chapter:

- H_0 always contains equality, such as $=$;

- H_1 uses one of these symbols: $<$, $>$, or $\neq$.

4. Writing Hypotheses from a Claim

Steps for Writing H_0 and H_1

1. Identify the claim and write it in symbolic form.
2. Write the symbolic form that must be true when the claim is false.
3. The statement without equality becomes H_1.
4. The null hypothesis H_0 uses equality with the fixed value.

Example 1: Given the claim that most Internet users utilize two-factor authentication, write the null and alternative hypotheses.

Solution

The claim is

$$p > 0.5.$$

If this claim is false, then

$$p \leq 0.5.$$

The alternative hypothesis is the statement that does not contain equality:

$$H_1 : p > 0.5.$$

The null hypothesis uses equality:

$$H_0 : p = 0.5.$$

So the hypotheses are

$$\boxed{H_0 : p = 0.5} \quad \text{and} \quad \boxed{H_1 : p > 0.5}.$$

5. Significance Level

The significance level, denoted by α, is the probability used as a cutoff for deciding whether the sample evidence is significant.

$$\alpha = P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true}).$$

Common choices are

$$\alpha = 0.05, \qquad \alpha = 0.01, \qquad \alpha = 0.10.$$

The most common choice is

$$\alpha = 0.05.$$

Interpretation of $\alpha = 0.05$

A significance level of 0.05 means that we are using a 5% cutoff for deciding whether the sample result is unusual enough to reject H_0.

6. Test Statistic

A test statistic is a value used to make a decision about the null hypothesis.

It is found by converting a sample statistic to a standardized score, assuming that H_0 is true.

Examples include

$$z, \qquad t, \qquad \chi^2.$$

Which Distribution Do We Use?

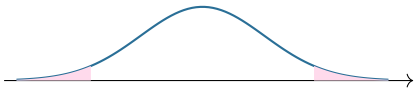
Claim About	Common Test Distribution
Population proportion p	Standard normal z
Population mean μ	Student t or standard normal z
Population standard deviation σ or variance σ^2	Chi-square χ^2

7. Types of Tests

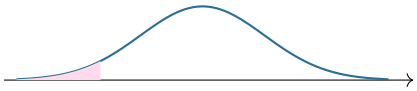
The direction of the alternative hypothesis determines the type of test.

Left-Tailed, Right-Tailed, and Two-Tailed Tests

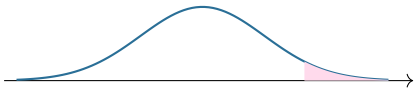
- If H_1 uses $<$, the test is left-tailed.
- If H_1 uses $>$, the test is right-tailed.
- If H_1 uses $\neq$, the test is two-tailed.



two-tailed



left-tailed



right-tailed

For the two-factor authentication claim,

$$H_1 : p > 0.5,$$

so the test is a right-tailed test.

8. P-Value Method

The P-value is the probability of getting a test statistic at least as extreme as the one obtained from the sample data, assuming that H_0 is true.

Decision Rule for the P-Value Method

If P-value $\leq \alpha$, reject H_0.

If P-value $> \alpha$, fail to reject H_0.

A helpful memory phrase is:

If the P is low, the null must go.

Example 2: For the two-factor authentication example, technology gives

$$z = 1.25$$

and

$$\text{P-value} = 0.106.$$

Using $\alpha = 0.05$, make a decision.

Solution

Compare the P-value with α:

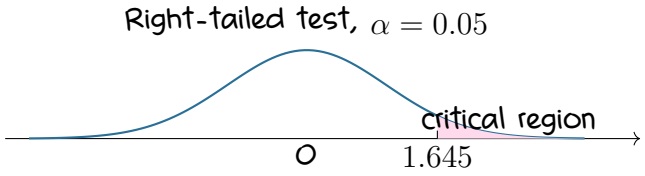
$$0.106 > 0.05.$$

Because the P-value is greater than the significance level, we fail to reject H_0.

9. Critical Value Method

With the critical value method, we compare the test statistic to one or more critical values.

The critical region is the area corresponding to test statistic values that cause us to reject H_0.



For a right-tailed test with $\alpha = 0.05$, the critical value is

$$z = 1.645.$$

Example 3: For the two-factor authentication example, the test statistic is

$$z = 1.25.$$

Using the critical value method with $\alpha = 0.05$, make a decision.

Solution

For this right-tailed test, the critical region begins at

$$z = 1.645.$$

The test statistic is

$$z = 1.25,$$

which is not in the critical region. Therefore, we fail to reject H_0.

10. Final Conclusion

After making the decision, we must write a conclusion in simple language that addresses the original claim.

Do Not Say ``Accept H_0''

We usually say

$$\text{fail to reject } H_0$$

instead of

$$\text{accept } H_0.$$

Failing to reject H_0 does not prove that H_0 is true. It only means the sample evidence is not strong enough to reject it.

Example 4: In the two-factor authentication example, we failed to reject

$$H_0 : p = 0.5.$$

Write the final conclusion about the original claim that most Internet users utilize two-factor authentication.

Solution

Because we failed to reject H_0, we do not have enough evidence to support the original claim.
A good final conclusion is:

There is not sufficient evidence to support the claim that most Internet users utilize two-factor authentication to protect their online data.

11. Confidence Interval Method

A confidence interval contains likely values of a population parameter.

Therefore, we can sometimes use a confidence interval for a hypothesis test.

Confidence Interval Idea

If the claimed value is not contained in the confidence interval, we reject the claim.

Significance Level	Two-Tailed Test	One-Tailed Test
$\alpha = 0.01$	99% CI	98% CI
$\alpha = 0.05$	95% CI	90% CI
$\alpha = 0.10$	90% CI	80% CI

Important Caution

For hypothesis tests about proportions, the confidence interval method might not always give exactly the same conclusion as the P-value method.

12. Type I and Type II Errors

Because hypothesis tests are based on sample data, mistakes are possible.

Type I Error

A Type I error is the mistake of rejecting H_0 when H_0 is actually true.

$$\alpha = P(\text{Type I error}).$$

Type II Error

A Type II error is the mistake of failing to reject H_0 when H_0 is actually false.

$$\beta = P(\text{Type II error}).$$

	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

Example 5: Consider the claim that a medical procedure increases the likelihood of a baby girl.

$$H_0 : p = 0.5$$

$$H_1 : p > 0.5$$

Describe a Type I error and a Type II error.

Solution

A Type I error is rejecting a true null hypothesis.
Here, that means concluding that the medical procedure is effective when in reality

$$p = 0.5,$$

so the procedure has no effect.
A Type II error is failing to reject a false null hypothesis.
Here, that means failing to conclude that the procedure is effective when in reality

$$p > 0.5,$$

so the procedure really does increase the likelihood of a baby girl.

13. Power of a Hypothesis Test

The power of a hypothesis test is the probability of rejecting a false null hypothesis.

$$\text{Power} = 1 - \beta.$$

Meaning of Power

A test with high power is more likely to detect a real effect when the effect truly exists.

For example, if the true value of the population proportion is farther from the null hypothesis value, the test usually has more power.

Example 6: A test of

$$H_0 : p = 0.5 \qquad H_1 : p > 0.5$$

uses $\alpha = 0.05$. Suppose technology gives the following powers for different true values of p.

True value of p	β	Power
0.6	0.820	0.180
0.7	0.564	0.436
0.8	0.227	0.773
0.9	0.012	0.988

What happens to power as the true value of p moves farther away from 0.5?

Solution

The power increases.
When $p = 0.6$, the power is only 0.180. When $p = 0.9$, the power is 0.988.
This makes sense because a larger difference from the null value $p = 0.5$ is easier to detect.

14. Summary of the Hypothesis Testing Process

Eight General Steps

1. Identify the claim and write it in symbolic form.
2. Write the symbolic form of the opposite statement.
3. Identify H_0 and H_1.
4. Select the significance level α.
5. Identify the appropriate test statistic and distribution.
6. Find the test statistic and P-value or critical value.
7. Reject H_0 or fail to reject H_0.
8. Write the conclusion in simple language.

Key Ideas

Important Takeaways

- A hypothesis test is used to test a claim about a population parameter.
- The null hypothesis H_0 always uses equality.
- The alternative hypothesis H_1 uses $<$, $>$, or $\neq$.
- The P-value method compares the P-value to α.
- The critical value method compares the test statistic to the critical region.
- We say ``fail to reject $H_0$,'' not ``accept H_0.''

- A Type I error rejects a true H_0.
- A Type II error fails to reject a false H_0.
- Power is the probability of rejecting a false null hypothesis.